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$$\begin{aligned} & 3z^5 + 2z^4 + 24z^3 + 16z^2 - 27z - 18 \\ \Leftrightarrow & z^2(3z^3 + 2z^2 + 24z + 16) - 9(3z + 2) \\ \Leftrightarrow & z^2(3z + 2) + 8(3z + 2) \\ \Leftrightarrow & (3z + 2)(z^2 + 8) \\ \Leftrightarrow & (3z + 2)(z^2(z^2 + 8) - 9) = (3z + 2)(z^4 + 8z^2 - 9) \\ \Leftrightarrow & z^4 + 8z^2 - 9 = u^2 + 8u - 9 = (z - 1)(z + 1) \\ \Leftrightarrow & A(z) = (3z + 2)(z - 1)(z + 1)(z^2 + 9) \\ z = & \frac{-2}{3} \vee z = 1 \vee z = -1 \vee z = 3i \vee z = -3i \end{aligned}$$

References